MTH108 2021S2 Exam

Feel free to contact me at maths_and_programming@outlook.sg if you need tuition. I provide university level tuition for the following SUSS units:

*	ENG301	Microprocessor Programming
*	ICT133	Structured Programming
*	ICT162	Object Oriented Programming
•	ICT235	Data Structures and Algorithms
•	ICT325	Algorithm Design and Analysis
•	ICT330	Database Management Systems
*	MTD215	Application of C++ in Multimedia
*	MTD315	Computer Interactive Graphics
*	MTH105	Fundamentals of Mathematics
*	MTH109	Calculus
•	MTH207	Linear Algebra
*	MTH210	Fundamentals of Probability
*	MTH212	Statistical Analysis
•	MTH240	Engineering Mathematics 1
*	MTH316	Multivariable Calculus
*	MTH321	Engineering Mathematics 2
*	MTH355	Basic Mathematical Optimisation

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SINGAPORE UNIVERSITY OF SOCIAL SCIENCES

MTH108

Timed Online Assignment – July Semester 2021

Calculus II

Tuesday, 23 November 2021 **4:00 pm – 6:30 pm**

Time allowed: 2.5 hours

INSTRUCTIONS TO STUDENTS:

1. This Timed Online Assignment (TOA) contains **FIVE (5)** questions and comprises **FOUR (4)** pages (including cover page).

2. You must answer **ALL** questions.

3. If you have any queries about a question, or believe there is an error in the question, briefly explain your understanding and assumptions about that question before attempting it.

4. You MUST **submit your answers via Canvas (similar to TMA submission) at the end time of this TOA** (as stated on this cover page). The 15 minutes grace period as shown on Canvas is strictly meant for technical issues encountered during submission. Thereafter, you will not be able to submit your answers and you will be considered as having withdrawn from the course. **No appeal will be allowed.**

5. You are to include the following particulars in your submission: TOA Course Code, your Full Name, and SUSS PI Number. Name your submission file as TOACourseCode_FullName_StudentPI. Use underscore and NOT Space. *Example: TOAXYZ_RaphaelLee_T1923161 (omit D/O, S/O).*

6. Your submission should consist of only one file and **must not exceed 500MB** in size. The file must be a PDF or Microsoft Word file saved in .doc or .docx format. All answers are to be typed. Flowcharts and graphs may be scanned or photographed and embedded in the PDF or Word file provided it does not exceed the file size limit of 500MB.

7. For answers which cannot be typed and require to be hand-written, ensure that question number is clearly stated on each page. All uploaded hand-written answers must be clear, readable and complete. Marks will not be awarded for un-readable or incomplete images.

8. To prevent plagiarism and collusion, your submission will be reviewed by Turnitin. The Turnitin report will only be made available to the marker and you will not be able to view it.

9. The University takes plagiarism and collusion seriously, and your Turnitin report will be examined thoroughly as part of the marking process.

10. Marks will only be awarded if FULL working is shown.

Answer all questions. (Total 100 marks)

Question 1

(a) Let $f(x) = e^{x-2} - 2$, defined over the domain $x \leq 1$. Determine an expression for $f^{-1}(x)$, as well as its domain and range.

(7 marks)

(b) Determine the derivative of $g(x) = \sqrt{1-x^2}\,\sin^{-1}(x)$ for $-1 < x < 1$, simplifying your answer as much as possible.

(6 marks)

(c) For the function $h(x) = \log_{2021}\left(\frac{x^{2021}}{2021^x}\right)$, determine the exact value of $h'(2021)$.

(7 marks)

Question 2

Use various techniques of integration to evaluate the following integrals.

(a) $\int \frac{\cos x}{(\sin x)^2 + 3\sin x}\,dx$

(8 marks)

(b) Determine the exact value of $\int_1^2 3xe^x\,dx$.

(8 marks)

Question 3

(a) Interpret $\lim\limits_{n\to\infty} \sum_{k=1}^{n} \frac{\frac{5}{4n}}{1+\frac{5k}{4n}}$ as a definite integral and evaluate it.

(8 marks)

(b) Let $f(x) = \int_1^{x^2} \frac{1+\sin t}{\sqrt{t}}\,dt$. Solve for $f'(x)$ and show that $f'(x) \leq 4$ for every $x > 0$.

(8 marks)

(c) Show that the following reduction formula holds, by using integration by parts:

$$\int \frac{e^{mx}}{x^n}\,dx = -\frac{e^{mx}}{(n-1)\,x^{n-1}} + \frac{m}{n-1}\int \frac{e^{mx}}{x^{n-1}}\,dx\,,$$

where $m, n \in \mathbb{Z}$ and $n > 1$.

(8 marks)

Question 4

Let R be the region in the first quadrant bounded by the curves $y = x^2 + 1$ and $y = 1 + 4x - x^2$. [A sketch of the region might be helpful.]

(a) Find the area of the region R.

(6 marks)

(b) Using the disk/washer method, find the volume of the solid of revolution obtained when R is rotated 2π radians about the x-axis.

(7 marks)

(c) Using the method of cylindrical shells, find the volume of the solid of revolution obtained when R is rotated 2π radians about the line $x = -1$.

(7 marks)

Question 5

(a) Evaluate the following integral, showing all steps of your working clearly:

$$\int \tan^{-1} x \; dx.$$

(8 marks)

(b) Hence, or otherwise, determine the exact value of

$$\int_0^1 (\tan^{-1} x)\left(\frac{x^2}{1+x^2}\right) dx.$$

(12 marks)

----- END OF PAPER -----

1(a) $f(x) = e^{x-2} - 2$

domain of $f(x) = \{x : x \leq 1\}$

$\max f(x) = f(1)$ since $f(x)$ is an increasing function

$= e^{-1} - 2$

$\min f(x) = \lim_{x \to -\infty} f(x)$

$= \lim_{x \to -\infty} (e^{x-2} - 2)$

$= 0 - 2$

$= -2$

range of $f(x) = (-2, e^{-1} - 2]$

let $y = e^{x-2} - 2$

$y + 2 = e^{x-2}$

$x - 2 = \ln(y+2)$

$x = 2 + \ln(y+2)$

$\therefore f^{-1}(x) = 2 + \ln(x+2)$

domain of f^{-1} = range of f = $(-2, e^{-1} - 2]$

range of f^{-1} = domain of f = $(-\infty, 1]$

(b) $g(x) = \sqrt{1-x^2}\,\sin^{-1} x$

$g'(x) = \frac{1}{2}(1-x^2)^{-\frac{1}{2}}(-2x)\sin^{-1} x + \sqrt{1-x^2}\left(\frac{1}{\sqrt{1-x^2}}\right)$

$= \frac{-x\sin^{-1} x}{\sqrt{1-x^2}} + 1$

(c) $h(x) = \log_{2021} x^{2021} - \log_{2021} 2021^x$

$= 2021 \log_{2021} x - x$

$= \frac{2021}{\ln 2021} \ln x - x$

$h'(x) = \frac{2021}{\ln 2021} \cdot \frac{1}{x} - 1$

$\therefore h'(2021) = \frac{2021}{\ln 2021} \cdot \frac{1}{2021} - 1$

$= \frac{1}{\ln 2021} - 1$

2(a) let $u = \sin x$

$du = \cos x \, dx$

$$\int \frac{\cos x}{\sin^2 x + 3\sin x} \, dx$$

$$= \int \frac{1}{u^2 + 3u} \, du$$

$$= \int \frac{1}{u(u+3)} \, du$$

$$= \frac{1}{3} \int \frac{1}{u} - \frac{1}{u+3} \, du$$

$$= \frac{1}{3}\left[\ln u - \ln(u+3)\right] + C$$

$$= \frac{1}{3} \ln \frac{u}{u+3} + C$$

$$= \frac{1}{3} \ln \frac{\sin x}{\sin x + 3} + C$$, where C is a constant

let $\frac{1}{u(u+3)} = \frac{A}{u} + \frac{B}{u+3}$

$\Leftrightarrow 1 = A(u+3) + Bu$

$= (A+B)u + 3A$

$\Leftrightarrow A+B = 0, \; 3A = 1$

$A = \frac{1}{3}$

$B = -A = -\frac{1}{3}$

$$\therefore \frac{1}{u(u+3)} = \frac{1}{3}\left[\frac{1}{u} - \frac{1}{u+3}\right]$$

(b) $\int \underbrace{3x}_{u} \underbrace{e^x dx}_{dv} = 3xe^x - \int 3e^x \, dx$

$= 3xe^x - 3e^x + C$

$= 3e^x(x-1) + C$ where C is a constant

$$\int_1^2 3xe^x \, dx = \left[3e^x(x-1)\right]_1^2$$

$$= 3e^2$$

3(a) $$\lim_{n\to\infty} \sum_{k=1}^{n} \frac{\frac{5}{4n}}{1+\frac{5k}{4n}} = \lim_{n\to\infty} \frac{1}{1+\underbrace{k\left(\frac{5/4}{n}\right)}_{x_k = k\Delta x}} \cdot \underbrace{\frac{\frac{5}{4}}{n}}_{\Delta x} \qquad f(x) = \frac{1}{1+x}$$

$$= \int_0^{\frac{5}{4}} \frac{1}{1+x}\,dx$$

$$= \left[\ln(x+1)\right]_0^{5/4}$$

$$= \ln\frac{9}{4}$$

$$= \ln 9 - \ln 4$$

(b) let $g(t) = \dfrac{1+\sin(t)}{\sqrt{t}}$

$$f(x) = \int_1^{x^2} g(t)\,dt = G(x^2) - G(1)$$

$$f'(x) = \frac{d}{dx}\left[G(x^2) - G(1)\right]$$

$$= \frac{d}{dx} G(t_2) \quad \text{where } t_2 = x^2$$

$$= \frac{dG}{dt_2} \cdot \frac{dt_2}{dx}$$

$$= g(t_2) \cdot \frac{dt_2}{dx}$$

$$= g(x^2) \cdot (2x)$$

$$= \frac{2x(1+\sin x)}{x}$$

$$= 2(1+\sin x)$$

when $x > 0$,

$-1 \le \sin x \le 1$

$0 \le 1+\sin x \le 2$

$0 \le 2(1+\sin x) \le 4$

$\therefore f'(x) \le 4$

(c) $$\int \frac{e^{mx}}{x^n}\,dx = \int \underbrace{e^{mx}}_{u}\,\underbrace{x^{-n}\,dx}_{dv} \qquad \text{using integration by parts}$$

$$= e^{mx}\cdot\frac{x^{-(n-1)}}{-(n-1)} - \int m e^{x} \cdot \frac{x^{-(n-1)}}{-(n-1)}\,dx$$

$$= -\frac{e^{mx}}{(n-1)\,x^{n-1}} + \frac{m}{n-1}\int \frac{e^{x}}{x^{n-1}}\,dx$$

5(a) $\int \tan^{-1} x = x\tan^{-1} x - \int \frac{x}{1+x^2}\,dx$

$= x\tan^{-1} x - \frac{1}{2}\int \frac{2x}{1+x^2}\,dx$

$= x\tan^{-1} x - \frac{1}{2}\ln(x^2+1) + C$, where C is a constant

(b) $\int (\tan^{-1} x)\left(\frac{x^2}{x^2+1}\right) dx$

$= \int (\tan^{-1} x)\left(\frac{x^2+1-1}{x^2+1}\right) dx$

$= \int (\tan^{-1} x)\left(1 - \frac{1}{x^2+1}\right) dx$

$= \int \tan^{-1} x - \frac{\tan^{-1} x}{x^2+1}\,dx$

$= \int \tan^{-1} x\,dx - \frac{1}{2}\int 2\left(\frac{1}{x^2+1}\right)\tan^{-1} x\,dx$

$= x\tan^{-1} x - \frac{1}{2}\ln(x^2+1) - \frac{1}{2}(\tan^{-1} x)^2 + C$, where C is constant

$\int_0^1 (\tan^{-1} x)\left(\frac{x^2}{x^2+1}\right) dx = \left[x\tan^{-1} x - \frac{1}{2}\ln(x^2+1) - \frac{1}{2}(\tan^{-1} x)^2\right]_0^1$

$= \left(\frac{\pi}{4} - \frac{1}{2}\ln 2 - \frac{\pi^2}{32}\right) - 0$

$= \frac{\pi}{4} - \frac{1}{2}\ln 2 - \frac{\pi^2}{32}$